Sheet (3)

(1) A regular polygon is an n-sided polygon in which all sides are of the same length and all angles have the same degree (i.e., the polygon is both equilateral and equiangular). The formula for computing the area of a regular polygon is

$$area = \frac{n \times s^2}{4 \times \tan\left(\frac{\pi}{n}\right)}$$

Here, s is the length of a side. Write a C++ program that prompts the user to enter the number of sides and their length of a regular polygon and displays its area.

(2) If the sides of a triangle are a, b and c then its angles can be expressed as;

$$A = \cos^{-1}\left(\frac{a^2 - b^2 - c^2}{-2bc}\right),\ B = \cos^{-1}\left(\frac{b^2 - a^2 - c^2}{-2ac}\right),\ C = \cos^{-1}\left(\frac{c^2 - b^2 - a^2}{-2ab}\right)$$

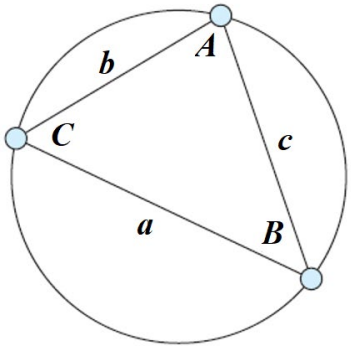
Write a C++ program, which prompts the user to enter the three sides of a triangle and calculate the corresponding angles. Check that the three sides give a valid triangle.

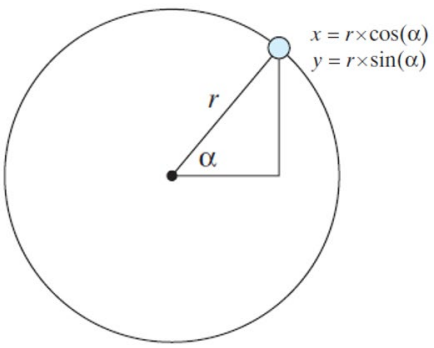
(3) The great circle distance is the distance between two points on the surface of a sphere. Let (x_1, y_1) and (x_2, y_2) be the geographical latitude and longitude of two points. The great circle distance between the two points can be computed using the following formula:

$$d = R.\cos^{-1}\left[\sin x_1 \sin x_2 + \cos x_1 . \cos x_2 . \cos\left(y_1 - y_2\right)\right]$$

Write a C++ program that prompts the user to enter the latitude and longitude of two points on the earth in degrees and displays its great circle distance. The average earth radius is $R = 6{,}378.1\ km$. The latitude and longitude degrees in the formula are for north and west. Use negative to indicate south and east degrees.

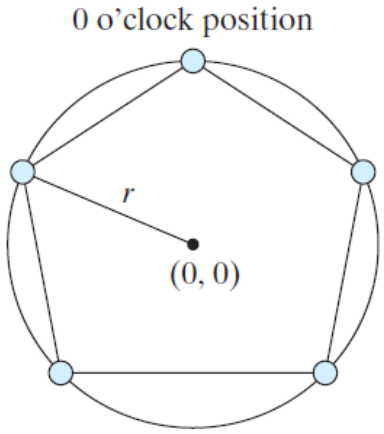
(4) Write a C++ program that generates three random points on a circle centered at $(0, 0)$ with radius 40 and display three angles in a triangle formed by these three points, as shown in in the below Figure.





(Hint: Generate a random angle a in radians between 0 and 2π, and the point determined by this angle is $\big(r \times cos(\alpha)\,,\ r \times sin(\alpha)\big)$.

(5) Suppose a pentagon is centered at $(0, 0)$ with one point at the 0 o'clock position, as shown in the below figure. Write a C++ program that prompts the user to enter the radius of the bounding circle of a pentagon and displays the coordinates of the five corner points on the pentagon.



(6) Write a C++ program to find the largest n such that $2^n < 30000$. Use a while loop to find the largest integer n such that 2^n is less than 30,000.

(7) Write a C++ program to calculate the product of the following series:

$$\frac{1}{5}\ ,\ \frac{1}{9},\ \frac{1}{13}\ ,\ \frac{1}{17}\ ,\ \frac{1}{21}\ ,\ \frac{1}{25}\ ,\ \cdots\ ,\ \frac{1}{93}\ ,\ \frac{1}{97}$$

(8) The ratio π can be calculated from *Leonhard Euler's* formula which is expressed as follows; $\frac{\pi^2}{6} = \sum_{n=1}^{\infty} \frac{1}{n^2}$. Therefore π is expressed as

$$\pi = \sqrt{6 \times \left(1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \ldots + \frac{1}{n^2}\right)}$$

Write a C++ program to calculate an approximate values of π and displays these values for $n = 100, 200,\ \ldots\ and\ 600$.

(9) The exponential function is expressed by *Taylor* series as $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ which can be rewritten as; $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ \ldots\ + \frac{x^n}{n!}$

Write a C++ program that prompts the user to enter x and then calculate and displays the value of e^x for $n = 15$.

(**Hint:** Because $n! = n \times (n-1) \times \ldots \times 2 \times 1$ then $\frac{1}{n!} = \frac{1}{n(n-1)!}$. Initialize e^x and item to be 1 and keep adding a new item to e^x. The new item is the previous item multiplied by x and divided by n for $n = 2,\ 3,\ 4, \ldots,\ 15$.

(10) Write a C++ program to find the greatest common divisor (GCD) of two integers n_1 and n_2 is as follows: First find d to be the minimum of n_1 and n_2, then check whether $d,\ d-1,\ d-2, \ldots, 2$, or 1 is a divisor for both n_1 and n_2 in this order. The first such common divisor is the greatest common divisor for n_1 and n_2. Write a program that prompts the user to enter two positive integers and displays the GCD.